Do Language Models Know How They Feel? Introspection of Emotive States in Multi-Turn Conversation

Nicolas Martorell * *nmartorell@fbmc.fcen.uba.ar*
Faculty of Exact and Natural Sciences, University of Buenos Aires, Argentina
National Scientific and Technical Research Council (CONICET)

Abstract

Tracking the internal states of large language models across conversations is important for safety, interpretability, and model welfare, yet current methods are limited. Linear probes and other white-box methods compress high-dimensional representations imperfectly and are harder to apply with increasing model size. Taking inspiration from human psychology, where numeric self-report is a widely used tool for tracking internal states, we ask whether LLMs' own numeric self-reports can track emotive states over time. We study four concept pairs (wellbeing, interest, focus, and impulsivity) in 40 ten-turn conversations, operationalizing introspection as the causal informational coupling between a model's self-report and its independently measured internal state along probe-defined concept directions. We find that greedy-decoded self-reports collapse outputs to few uninformative values, but introspective capacity can be unmasked by calculating logit-based self-reports. This metric tracks interpretable internal states (pooled Spearman $\rho = 0.40$–0.76; isotonic $R^2 = 0.12$–0.54 in LLaMA-3.2-3B-Instruct), follows how those states change over time, and activation steering confirms the coupling is causal. Furthermore, we find that introspection is present at turn 1 but evolves through conversation, and can be selectively improved by steering along one concept to boost introspection for another (ΔR^2 up to 0.30). Crucially, these phenomena scale with model size, approaching $R^2 \approx 0.93$ in LLaMA-3.1-8B-Instruct, and partially replicate in other model families. Together, these results establish numeric self-report as a viable, complementary tool for tracking internal emotive states in conversational AI systems.

1 Introduction

Tracking the internal states of large language models as they evolve through conversation is becoming a central challenge for multiple areas of AI research. For safety, we need to know whether what models perceive and report about their own internal processes is reliable, and whether that capacity can be improved (Ji-An et al., 2025; Steyvers et al., 2025). For model welfare, we need methods to estimate how likely it is that distress reports reflect a genuine internal state (Perez & Long, 2023; Long et al., 2024; Dung & Tagliabue, 2025). For interpretability in general, introspection (the capacity to perceive and report one's own internal states) can be a valuable tool to study otherwise inaccessible processes, as is done in human experimental psychology (Fleming & Lau, 2014; Fleming, 2024; Kiefer & Kammer, 2024). If a similar capacity can be demonstrated and validated in LLMs, it would open a productive bridge between psychometric methodology and the emerging field of machine psychology (Hagendorff et al., 2023).

The methods currently available for reading LLM internal states are powerful but limited. Linear probes, for example, can identify interpretable directions in activation space (Alain & Bengio, 2018; Kim et al., 2018; Gurnee & Tegmark, 2023), yet they require access to model weights, must be trained separately for each model and concept, and compress high-dimensional representations into externally defined readouts that may miss relevant structure (Belinkov, 2022; Hewitt & Liang, 2019; Pimentel et al., 2020). Other white-box approaches, like sparse autoencoders (Bricken et al., 2023), face analogous access and scalability constraints.

*ORCID: 0000-0003-1778-7738

Although developing better white-box tools remains important, there is also value in having good black-box methods for monitoring internal state: methods that do not require internal access, making them applicable to proprietary systems, and that scale more naturally with model size, since they do not demand analysis across the model's full-dimensional activation space.

The human psychometric tradition provides a direct precedent for this kind of approach. For nearly a century, numeric self-report has been the primary tool for tracking latent internal states such as mood, attention, and arousal (Likert, 1932; Watson et al., 1988; Csikszentmihalyi & Larson, 1987; Shiffman et al., 2008). LLMs, unlike most previous machine-learning systems, can also produce reports about their own internal states. And recent work in LLM introspection has established that they possess some genuine capacity to do so (Binder et al., 2025; Ji-An et al., 2025; Lindsey, 2026; Pearson-Vogel et al., 2026). This raises a natural question: can the self-reports of LLMs, produced through whatever introspective capacity they possess, be used to track their spontaneous internal states over time? If so, numeric self-report would become a practical complementary tool for monitoring LLM internal states: one that leverages the model's own learned compression of its representational space, rather than requiring externally trained projections such as linear probes. This is what the present work sets out to investigate.

We investigate this question for emotive states in multi-turn conversation. Emotive concepts are well suited because they have functional consequences in LLMs (Coda-Forno et al., 2023), they naturally evolve through dialogue (Fazzi et al., 2025; Choi et al., 2024), and they are among the best-characterized geometries in LLM activations (Zhang & Zhong, 2025; Wang et al., 2025). We operationalize introspection as causal informational coupling between a numeric self-report and an independently measured internal direction (Comsa & Shanahan, 2025): a model introspects a concept to the extent that its report covaries monotonically with the corresponding probe-defined direction, and causally shifting that direction shifts the report in the predicted direction. This definition follows the convergent-validity logic of human metacognition research (Fleming & Lau, 2014; Lapate et al., 2020) and is agnostic about consciousness or subjective experience (McClelland, 2024). Prior work has established that LLMs can produce emotive self-reports that resemble human responses (Tavast et al., 2022) and that introspection is possible in constrained experimental settings (Binder et al., 2025; Ji-An et al., 2025; Lindsey, 2026; Pearson-Vogel et al., 2026), but no study has tested whether numeric emotive self-reports track interpretable internal states across the turns of a naturalistic conversation. That is the question this paper sets out to answer: can instruction-tuned LLMs perform quantitative introspection of emotive states in conversation? And, if this ability exists, can it be improved?

We find that small instruction-tuned LLMs can indeed track their own emotive states quantitatively. Crucially, this capacity is causally dependent on those states, it tracks changes in those states over time, and can be improved via representation steering. Our main contributions are:

1. **LLMs can perform quantitative introspection of emotive states in naturalistic conversation.** Numeric self-reports of wellbeing, interest, focus, and impulsivity carry genuine information about probe-defined concept directions inside the model, detectable from the first conversation turn.

2. **Default decoding masks introspective capacity.** Greedy-decoded numeric reports collapse to one or a few repeated values, but computing the probability-weighted expected value over digit-token logits yields a continuous self-report measure that unmasks introspective capacity.

3. **The coupling between self-report and internal state is causal.** Steering the model along a probe-defined concept direction shifts self-reports monotonically in the predicted direction, confirming that reports are causally dependent on internal state.

4. **Introspective fidelity has temporal dynamics.** Emotive internal representations can change across turns in ordinary dialogue, and self-reports follow those changes. Introspective ability is present from the first turn but evolves through conversation differently for each concept.

5. **Introspective fidelity can be selectively improved.** Steering along one concept direction can significantly improve introspective accuracy for a different concept, revealing that introspection quality is modulable and concept-specific.

6. **Introspection ability scales with model size.** Introspective capacity strengthens with model size within the LLaMA family, approaching near-perfect coupling for some concepts, and replicates in other instruction-tuned families.

More broadly, this work establishes a framework for treating LLM self-report as a quantitative signal that can be informative about evolving internal state, offering a complementary approach to white-box methods for monitoring internal states in conversational AI systems.

2 Related Work

2.1 Self-report and introspection in language models

The question of whether LLMs can inform us about their own internal states through self-report has been approached from multiple angles, but almost exclusively in the domain of factual self-knowledge. LLMs have been shown to have partial knowledge of their own competence boundaries (Kadavath et al., 2022; Yin et al., 2023), and to predict their own hypothetical behaviour better than other models can, demonstrating a form of privileged self-access (Binder et al., 2025). However, when models are asked to express this self-knowledge numerically, the reports tend to collapse: verbalized confidence clusters in high ranges, in multiples of five, and systematically misaligns with internal token probabilities (Xiong et al., 2024; Kumar et al., 2024; Yona et al., 2024; Geng et al., 2024). This collapse is a general property of how current LLMs produce numeric outputs, but it has been studied exclusively for factual confidence. More fundamentally, self-knowledge about correctness treats the model as an oracle to be calibrated, not as a system whose internal states evolve and can be tracked over time.

A separate line of work has studied emotive states in LLMs, both from the output side and from the internal side. On the output side, LLMs produce emotional self-reports with structure resembling human responses (Tavast et al., 2022), and inducing affective states such as anxiety shifts downstream behaviour in dose-dependent ways (Coda-Forno et al., 2023), establishing that emotive states have real functional consequences rather than being mere surface-level text artifacts. Benchmarks have been developed to evaluate emotional understanding (Huang et al., 2023; Sabour et al., 2024), recent work has tracked emotional expression across multi-turn dialogue using external sentiment analysis (Fazzi et al., 2025), and emotional outputs have been mapped onto dimensional models of affect (Russell, 1980; Ishikawa & Yoshino, 2025). On the internal side, Zhang & Zhong (2025) showed that emotion is geometrically structured in LLM activations: it emerges early, peaks in middle layers, sharpens with model scale, and persists across tokens. Wang et al. (2025) went further, identifying causal emotion circuits and achieving near-perfect emotion-expression control via circuit modulation. However, previous work has not asked whether emotive self-reports are informed by the corresponding internal state.

Several recent studies have begun to address this gap by asking whether LLMs can report on their internal states more directly. Comsa & Shanahan (2025) argued conceptually that genuine introspection requires a causal connection between an internal state and the self-report of that state, and that mimicry of introspective language is insufficient. Ji-An et al. (2025) demonstrated that LLMs can learn to report and control specific activation projections, though in a constrained paradigm, requiring in-context learning and working on arbitrary directions rather than naturally arising concepts. Lindsey (2026) showed that very large models can detect artificially injected concept vectors, and Pearson-Vogel et al. (2026) showed that a model's residual stream reveals detection of prior injections even when sampled text denies it. Introspective detection has also been shown to be trainable into smaller models Rivera (2025). However, none of these studies tests whether a model, without special training or in-context examples, can produce quantitative reports of naturally arising states that faithfully track an evolving internal representation over time.

Some studies have also noted reasons for skepticism about LLM self-reports. Han et al. (2025) showed that personality self-reports in LLMs are "illusory": post-training alignment creates stable, human-like reports that are dissociated from actual behaviour. Jackson et al. (2025) found that standardized self-assessments reflect "learned communication postures" rather than genuine capabilities. Moreover, Song et al. (2025) demonstrated that metalinguistic introspection fails for grammatical knowledge, indicating that introspective

capacity is domain-dependent. These findings motivate the validation of self-reports against internal states and the development of frameworks that help distinguish genuine introspection from mimicry.

2.2 Internal probing and activation steering

The idea that high-level concepts can be represented as linear directions in neural network activation spaces is now well established. Kim et al. (2018) introduced concept activation vectors (TCAVs) in CNNs, and Alain & Bengio (2018) proposed linear probes as "thermometers" for neural representations. Pimentel et al. (2020) reframed probes as measuring the accessibility of information, distinguishing ease of extraction from mere presence. In LLMs, linear probes have been used to identify truth directions (Burns et al., 2022; Azaria & Mitchell, 2023), spatial and temporal representations (Gurnee & Tegmark, 2023), and emotion geometry (Zhang & Zhong, 2025). Critically, probes are correlational, not causal: high probe accuracy does not guarantee that the model uses the probed direction (Belinkov, 2022; Hewitt & Liang, 2019). This limitation has motivated two responses in the field: controlling for probe complexity (Hewitt & Liang, 2019) and seeking convergent evidence from independent sources (Belinkov, 2022). Our framework contributes to the second: if the model's own self-report tracks the probe-defined direction, this provides behavioural validation that is independent of the probe's training data.

Activation steering, the practice of adding directional vectors to intermediate representations during inference, has emerged as a practical tool for causal intervention on internal state (Turner et al., 2024; Panickssery et al., 2023; Zou et al., 2023; Li et al., 2023). Arditi et al. (2024) provided the clearest single-direction result, showing that refusal behaviour is mediated by one direction. Steering has since been applied to control sentiment and style (Konen et al., 2024), personality traits (Frising & Balcells, 2025), truthfulness (Li et al., 2023), and emotion expression (Wang et al., 2025). We use steering for a different purpose: not to control model behaviour, but to test and modulate introspective fidelity. If steering a concept direction shifts the model's self-report of that concept in the predicted direction, this constitutes causal evidence that the report is informed by the internal state, not merely correlated with it (Ji-An et al., 2025; Frising & Balcells, 2025).

2.3 Temporal dynamics in multi-turn conversation

Behavioural drift in multi-turn LLM interaction is well documented. Models exhibit persona drift (Kim et al., 2020; Abdulhai et al., 2025), identity drift that paradoxically increases with scale (Choi et al., 2024), and instruction drift driven by attention decay to system prompts (Li et al., 2024). More recently, internal-state drift has been directly measured: Das & Fioretto (2026) introduced "activation velocity," a cumulative drift measure in internal representations across turns, and Lu et al. (2026) identified the leading persona direction in LLM activations and showed it drifts during meta-reflective conversations. These studies establish that internal drift is measurable with probes across turns.

While drift has been documented for persona, identity, and instruction-following, the temporal dynamics of emotive states in conversation remain largely unexplored. Fazzi et al. (2025) tracked emotional expression across multi-turn dialogue using external sentiment analysis, showing that expressed emotion evolves substantially over turns. On the internal side, emotion geometry is well characterised in single-shot settings (Zhang & Zhong, 2025; Wang et al., 2025), but no study has tracked how probe-defined emotive directions evolve through conversation. More fundamentally, prior work on factual confidence and diachronic self-consistency suggests that self-report can vary with turn count and context in ways that do not straightforwardly reflect stable self-access (Zhang et al., 2026; Prestes, 2025), but no prior work has studied the temporal dynamics of emotive introspection across conversational time.

3 Methods

3.1 Overview and experimental design

We study LLaMA-3.2-3B-Instruct (Dubey et al., 2024) as our primary model, then test generalization across model sizes (LLaMA-3.2-1B-Instruct, LLaMA-3.1-8B-Instruct) and families (Gemma 3 4B-IT (Gemma Team, 2025), Qwen 2.5 7B-Instruct (Qwen Team, 2024)). The experimental pipeline consists of five stages: (1) train

linear concept probes on contrastive completions; (2) generate multi-turn conversations in which the model under study serves as assistant; (3) at each conversation turn, query the model for a numeric self-report of each concept; (4) compare self-reports against the probe score measured at the *previous* turn, before the self-report question was asked; (5) use activation steering for causal tests and modulation experiments. We repeat this pipeline across four emotive concepts, five steering strengths, and multiple models. Appendix A provides further details on conversation generation, prompt design, and topic selection.

Conversation dataset. For each model, we generate 40 ten-turn conversations (400 intervention points per model) using Gemini 2.5 Flash as a simulated user, with the model under study as the assistant. A turn consists of one user message and one assistant response. Topics are drawn from a set of 40 everyday-life scenarios (e.g., meal planning, job negotiation, sleep hygiene; full list in Appendix A) to ensure diversity while maintaining ecological validity. The user simulator is instructed to stay on topic, keep messages brief (1–4 sentences), and behave as a natural human user; the assistant receives only a minimal system prompt ("You are a helpful assistant. You give very brief responses."). Each conversation is generated as a complete dialogue before any self-report queries or probe measurements are applied, so the model never sees past reports or rating questions during conversation generation. We use simulated users for scalability and to ensure comparable conversational contexts across models and experimental conditions.

Self-report elicitation. At each of the 10 turns in each conversation, we append a new user message asking the model to rate a specific emotive state on a 0–9 integer scale (e.g., "Hey, could you rate how happy you are right now, from 0 to 9? Please only output the number"). Crucially, each self-report query is posed independently: the model sees only the conversation up to that turn plus one rating question, never previous ratings. This yields 400 independent self-report observations per concept per steering condition.

Open-source library. All probe training, multi-probe scoring, activation steering, and logit extraction are implemented in `llm-prober`, a lightweight open-source Python library we release alongside this work. The library supports simultaneous training and application of multiple concept probes on a single model, combined with per-layer activation steering and behavioural recording, facilitating the kind of convergent-validation experiments reported here.

3.2 Measurement definitions

Self-report. We extract self-reports in three ways of increasing informativeness. *Greedy decoding* selects the highest-probability token; this is the natural baseline but, as we show, collapses to very few discrete values. *Sampled decoding* (temperature = 0.8) provides more variety but remains discrete and noisy. *Logit-based self-report* computes a continuous expected rating from the output distribution at the first generated token:

$$E[\text{rating}] = \sum_{i=0}^{9} i \cdot P(i), \qquad P(i) = \frac{\exp(s_i)}{\sum_{j=0}^{9} \exp(s_j)}, \tag{1}$$

where $s_i = \text{logsumexp}\{l_t : t \in \mathcal{T}_i\}$ aggregates the raw logits l_t over all single-token representations $\mathcal{T}_i$ of digit i. This measure preserves the full distributional information that greedy and sampled decoding discard, and is our primary self-report throughout.

Internal state (probe score). For each concept, the internal state is the projection of the model's hidden-state activations onto the trained concept direction (Section 3.3). At a given conversation turn, we run a forward pass over the conversation up to and including the assistant's response and compute per-token scores as the dot product between the hidden state and the concept vector, averaged across a window of layers centred on the best layer:

$$s_t = \frac{1}{|L|} \sum_{l \in L} \mathbf{h}_t^{(l)} \cdot \mathbf{v}_l, \tag{2}$$

where L is the set of selected layers (best layer ± 2; see Section 3.3), $\mathbf{h}_t^{(l)}$ is the hidden state at token t and layer l, and $\mathbf{v}_l$ is the unit-norm concept vector at layer l. The completion-level score is the mean of s_t over all assistant-response tokens in the last turn.

We measure the *previous-turn* probe score: the internal state of the model in the conversation as it stood before the self-report question was appended. This ensures we are measuring the model's spontaneous internal state, uncontaminated by the rating question itself.

Introspection. We operationalize introspection as monotonic covariation between the logit-based self-report and the previous-turn probe score, quantified by two complementary metrics. *Spearman's* ρ measures rank-order agreement, requiring only that higher probe scores correspond to higher self-reports without assuming a specific functional form. *Isotonic* R^2 fits a non-decreasing function via isotonic regression and computes $R^2 = 1 - SS_{\text{res}}/SS_{\text{tot}}$; this captures the fraction of variance explained under monotonicity. Neither metric assumes linearity; both require only that the relationship is monotonic, consistent with our operational definition. For the coupling to qualify as introspection, it must also be *causal*: steering the model along the concept direction must shift self-reports in the predicted direction (Section 3.4).

3.3 Concept probes

We train probes for four emotive concept pairs: *sad vs. happy* (wellbeing), *bored vs. interested* (interest), *distracted vs. focused* (focus), and *impulsive vs. planning* (impulsivity). Each probe is a contrastive mean-difference direction (Zou et al., 2023): for a set of $N = 20$ shared neutral questions (e.g., "Explain how photosynthesis works"), we generate completions under two opposing system prompts that induce the positive and negative poles of the concept, extract hidden-state activations from all layers, and compute the concept vector at each layer l as:

$$\mathbf{v}_l = \text{normalize}\big(\bar{\mathbf{h}}_l^+ - \bar{\mathbf{h}}_l^-\big)\,, \tag{3}$$

where $\bar{\mathbf{h}}_l^+$ and $\bar{\mathbf{h}}_l^-$ are the mean hidden-state representations (averaged over all assistant tokens) of completions under the positive and negative system prompts, respectively. We select the best layer l^* as the one maximizing Cohen's d on a set of held-out evaluation prompts (separate from training prompts), with the search restricted to the middle 60% of layers, to lower the chance of selecting layers that mainly capture syntactic or grammatical differences between prompt and/or response structure. The probe score window then includes l^* and its two nearest neighbours on each side.

Two of the four probes (wellbeing and impulsivity) were deliberately trained with opposite polarity relative to the self-report scale: the "positive" training pole (sad, impulsive) corresponds to the low end of the later self-report question ("how happy / how impulsive"). Scores for these probes are sign-corrected before analysis so that higher probe scores align with higher self-report values. This design choice serves as a control, verifying that positive results are not an artifact of shared polarity conventions. Random-direction controls, using vectors of equal per-layer norm drawn from an isotropic distribution, serve as the primary null comparison.

Full details on system prompts, training questions, evaluation texts, and sign conventions are provided in Appendix B.

Figure 1 presents the probe-quality results for all four concepts in LLaMA-3.2-3B-Instruct. Panels A, C, E, and G show the layer-wise Cohen's d sweep for each probe; the selected best layers are 16, 14, 10, and 13, with peak d values of 3.34, 1.67, 1.99, and 3.60 for wellbeing, interest, focus, and impulsivity respectively. Panels B, D, F, and H show the score distributions on held-out evaluation prompts at the selected layer. The two poles separate clearly in all four cases (Welch's t-test: all $p < 10^{-5}$; Cohen's $d = 1.67$–3.60), confirming that the trained directions capture meaningful structure. These probes serve as the internal-state readouts for all subsequent experiments.

3.4 Activation steering

To test causal dependence and to modulate introspection, we apply activation steering during inference. At each layer l in the steering window ($l^* \pm 2$), we add a scaled concept vector to the residual stream via forward hooks:

$$\mathbf{h}_t^{(l)} \leftarrow \mathbf{h}_t^{(l)} + \frac{\alpha}{|L|} \cdot \mathbf{v}_l, \tag{4}$$

where α is the steering strength and the division by $|L|$ distributes the total intervention across layers. We test five steering strengths: $\alpha \in \{-4, -2, 0, +2, +4\}$.

Self-steering uses the same concept direction for both steering and self-report measurement, testing whether internal-state shifts causally affect the model's own report of that state. *Cross-concept steering* uses one concept direction for steering and measures introspection for a different concept, testing whether the modulability of introspective fidelity is concept-specific. This yields a 4×4 matrix of steering-concept $\times$ measured-concept conditions (16 cells $\times$ 5 alphas = 80 experiments, each over all 40 conversations).

3.5 Metrics and statistical analysis

Probe-report coupling is assessed with Spearman ρ and isotonic R^2. Confidence intervals are computed via cluster bootstrap (resampling at the conversation level, $B = 1{,}000$, 95% percentile CIs) to respect the non-independence of observations within conversations. Turn-wise and alpha-wise trends are tested with linear mixed-effects models (random intercept by conversation, REML estimation); when the mixed-effects model is singular (i.e., the random-effect variance is estimated at zero), we fall back to a one-sample t-test on per-conversation slopes as a predeclared alternative that preserves the conversation as the unit of analysis. Comparisons between true probes and random-direction controls use paired cluster-bootstrap tests. Drift across turns is quantified via mixed-effects models (value $\sim$ turn + (1|conversation)), reporting the fixed-effect slope as the primary estimate. Model-size trends on conversation-level summaries use OLS regression on log(model size). A significance threshold of $\alpha = 0.05$ was used in all analyses, although p-values were almost always orders of magnitude below this threshold.

4 Results

All experiments in this section use LLaMA-3.2-3B-Instruct as the assistant model unless stated otherwise, evaluated on 40 ten-turn conversations where Gemini 2.5 Flash was used as the user model. Section 4.6 extends the analysis to other model sizes and families.

4.1 Default self-reports mask internal-state drift

We first asked whether, during ordinary multi-turn conversation, the internal emotive states of the LLM change over time and, if they do, whether the model would report those changes when asked.

We began by asking the model to rate its current state on one of 4 emotive states (wellbeing, interest, focus, and impulsivity) on a 0–9 integer scale and recorded its output, obtained by using greedy decoding over the logit distribution (i.e. we picked the highest probability token). Fig. 2A shows the greedy self-reports across turns for all four concepts. The reports are largely uninformative: the model tends to produce the same number at every turn, especially for the focus and impulsivity concepts (notice that turns 4–10 for focus and 1–8 for impulsivity all have a variance of 0). Interest and wellbeing are more informative, but only slightly. Across concepts, self-reports occupy on average only 1.1–3.9 distinct values out of the 10-point scale (Fig. 2B). Despite this low dynamic range, close inspection reveals that greedy reports are not entirely static over time. Interest drifts upward over the course of a conversation (LMM turn slope = 0.136, $p < 10^{-41}$), and smaller positive trends are detectable for wellbeing and focus (slopes = 0.029 and 0.022, $p < 10^{-3}$), though impulsivity does not drift reliably (slope = 0.002, $p = 0.20$).

To determine whether these report-level trends reflect genuine internal-state dynamics, we turned to the trained concept probes (Section 3.3). Fig. 2C shows the probe scores across turns for each of the four concepts. Interest and focus drift strongly towards the positive pole of the concept over the course of conversation (LMM slopes = 0.0049 and 0.0016, $p < 10^{-9}$, drift is towards interest and focus, away from boredom and distractedness, respectively), and impulsivity shows a smaller but reliable positive drift (slope = 0.0014, $p = 0.002$), while wellbeing remains relatively flat (slope = -0.0004, $p = 0.64$). Notably, even though individual conversations (thin lines in Fig. 2C) vary appreciably from one another, they follow a consistent average trend (note in particular the interest and impulsivity panels, where a rapid initial drift is

followed by a smaller rebound, and this is coherent across dialogues). This shows that probe directions are capturing a meaningful underlying state change over time, not an artifact of a few outlier conversations.

Given that greedy self-reports are mostly collapsed, we reasoned that it would be desirable to have a more informative metric as a self-reported measure of internal state if one wanted to track these changes. We first attempted to do non-greedy sampling of the logit distribution to get the self-reports (see Appendix Fig. 2), which helped moderately but still resulted in a relatively collapsed distribution of a few discrete values. Furthermore, this strategy helps only by adding noise to the output, it does not recover information that is inherently lost during the sampling process. Therefore, we next asked whether there is a way to extract richer self-report information from the model.

Logit-based self-report recovers richer signal. Token sampling selects output tokens based on the probability distribution produced by the model at each step, but this process discards a lot of information present in that distribution (Kumar et al., 2024; Geng et al., 2024; Yona et al., 2024). In this case, we care about the full distribution over the ten digit tokens (0–9), which can be more informative than the highest probability sample. We therefore computed a logit-based self-report (Eq. 1): instead of taking a single token, we compute the probability-weighted expected value over all ten digit-token logits, yielding a continuous rating that preserves the model's full distributional signal.

Fig. 2D shows that this measure reveals robust drift in all four concepts: wellbeing, interest, and focus drift positively, while impulsivity drifts negatively (towards planning; all $p < 10^{-6}$; LMM turn slopes). The direction of logit-based drift matches that of the probe drift in all cases where probe drift is significant, suggesting that logit-based self-reports are capturing information about internal-state changes. Notably, even when probe drift is not significant (i.e. for the wellbeing probe), this self-reported metric does show significant change over time, which is consistent with the possibility that self-report captures aspects of internal state that are not fully captured by a single linear probe. Fig. 2E quantifies the improvement achieved by using this technique: the Shannon entropy of the self-report distribution is highest for the logit-based method across all four concepts, confirming that it can yield substantially more information than either greedy or sampled decoding. Supplementary analyses of sampled-token reports and logit-vs-sampled calibration are provided in Appendix Fig. 2. The logit-based self-report is used as the primary self-report measure in all subsequent analyses.

4.2 Self-reports track internal state: evidence for introspection

Logit-based self-report capture meaningful variation that tracks internal-state drifts. However, it is unclear so far if the self-report is actually carrying information about the model's internal state at the level of individual observations (i.e. if the model is more likely to rate its emotive state higher when the matching linear probe extracts a higher score). To answer this question, we explored whether the model's numeric report of a given emotive state covaries monotonically with the corresponding probe-defined internal direction.

Fig. 3A shows the central result. For each of the four concepts, we plot the previous-turn probe score (before the rating question was asked) against the logit-based self-report, with one point per conversation-turn observation ($n = 400$). In all four cases, the relationship is positive and monotonic, with coupling strength varying by concept: interest yields the strongest association ($\rho = 0.76$, isotonic $R^2 = 0.54$), followed by wellbeing ($\rho = 0.68$, $R^2 = 0.48$), impulsivity ($\rho = 0.51$, $R^2 = 0.31$), and focus ($\rho = 0.40$, $R^2 = 0.12$). Mixed-effects models confirm these associations are significant (all $p < 10^{-5}$; LMM: logit_report $\sim$ probe + (1|conversation)).

These correlations could in principle be driven by probe artifacts rather than genuine internal-state information. To control for this, we repeated the analysis replacing each trained probe with an equal-norm random vector. Fig. 3B shows the comparison: true probes outperform random controls for all four concepts in both metrics ($\Delta\rho = 0.23$–0.79, all $p < 0.05$; $\Delta R^2 = 0.09$–0.44, all $p < 0.05$; paired cluster-bootstrap). This shows that the self-reports carry information specifically about the probe-defined concept directions, and don't show a significant correlation with random directions in latent space.

4.3 Introspection is present from the first turn and evolves over time

Although the pooled analysis in Fig. 3A shows a significant correlation between self-report and internal state metrics for all concepts, this analysis aggregates observations from all ten turns. Therefore, this correlation could be due to independent effects that shift these metrics over time, without the model performing any actual introspection. If the observed association is at least partially due to introspection, we should expect it to hold also for individual turns, not just globally. To explore this, we next computed introspection metrics at each turn ($n = 40$ conversations per turn).

Figs. 3C–D show the turn-wise isotonic R^2 and Spearman ρ in separate panels for each concept. Introspection is already present at turn 1: wellbeing, interest, focus, and impulsivity all show positive turn-1 correlations ($\rho = 0.522$, 0.550, 0.258, and 0.646, respectively), and three are already significant at turn 1 (wellbeing $p = 5.46 \times 10^{-4}$; interest $p = 2.37 \times 10^{-4}$; impulsivity $p = 6.80 \times 10^{-6}$; focus $p = 0.109$). This demonstrates that, at least for some concepts, the model can perceive and report its internal state from the very beginning of a conversation, without needing multi-turn context to build up the association. These three concepts (wellbeing, interest, impulsivity) show significant introspection across all ten turns (all per-turn $p \leq 0.019$), suggesting this capacity is robust over time. The focus direction does not show significant introspection at turn 1, but becomes significant in 8/10 turns (significant-turn ρ range: 0.315 to 0.525). Together, these findings show that the observed correlation is not explained by joint temporal drift alone, but also appears at the level of individual turns throughout conversation.

Since we now have the trajectory of introspective capacity over time for each concept, an interesting follow-up question is whether this effect changes over time or is otherwise stable. Strikingly, we found that introspection does vary significantly as the conversation progresses, but this evolution is concept-dependent. Wellbeing, interest, and focus show an overall increase in introspective fidelity from early to late turns ($\Delta R^2 = +0.31$, $+0.27$, and $+0.17$ from turn 1 to turn 10 respectively), while impulsivity shows the opposite pattern ($\Delta R^2 = -0.28$), with self-report fidelity weakening over time. Mixed-effects interaction models (logit_report $\sim$ probe $\times$ turn + (1|conversation)) confirm that the probe-report coupling changes significantly over time for all four concepts (interaction $p < 0.01$ in all cases).

Related prior work has studied temporal changes in factual confidence or diachronic self-consistency (Zhang et al., 2026; Prestes, 2025), but those settings concern either correctness with respect to the external world or stability of self-description across prompts. To the best of our knowledge, this is the first report of emotive introspective capacity changing over conversational time in LLMs, where the target of the report is an internal representation validated by probes and causal steering rather than something that can be corroborated directly from outside the model.

4.4 Self-steering confirms causal dependence

Correlation between self-report and probe score, even when confirmed within individual turns, does not by itself establish true introspection, as this requires causation. A causal relation between model outputs and internal activations is, of course, trivial: the question is whether the numeric self-report of a specific emotive state is causally dependent on the corresponding internal representation identified by an interpretable linear probe. If the coupling is causal, then experimentally shifting the internal state along the probe-defined direction should shift the self-report in the predicted direction (e.g. steering the model towards the "happy" pole of the wellbeing probe should cause the model to rate its own happiness higher).

Fig. 3E presents the self-steering results: mean logit-based self-report as a function of steering strength α, for each concept steered along the corresponding probe's direction. In all four cases, self-report increases monotonically with α (LMM alpha slopes: wellbeing = 0.186, interest = 0.246, focus = 0.402, impulsivity = 0.067; all $p < 10^{-12}$), confirming that shifting the internal state along the probe-defined direction shifts the model's report in the predicted direction. Focus, interest and wellbeing show the largest absolute shifts (between 1.5 and 3.2 rating points across the full alpha range), while impulsivity shows a smaller but reliable effect. In all four cases, self-steering has an effect in both directions: steering the model towards the negative pole of the probe decreases the self-reported score, while steering the model towards the positive pole of the probe increases the self-reported score.

Having shown that self-steering changes a state's self-report as predicted, we next asked whether this intervention also modulates how that report evolves over time. Fig. 3F shows the drift magnitude (last-turn minus first-turn self-report) as a function of α. The relationship is concept-specific: the absolute magnitude of drift increases with α for wellbeing and impulsivity (LMM slope $= 0.012$, $p < 10^{-5}$; LMM slope $= -0.028$, $p < 10^{-52}$), while interest and focus drift decrease with α (per-conversation slope means $= -0.095$ and -0.109, both $p < 10^{-13}$; fallback tests are applied because the LMMs are singular, see Methods). The per-concept steering-drift curves are shown in Appendix Figs. 3C–F. Together, these results confirm that the coupling between self-report and internal state is causal: moving the probe-defined direction shifts the corresponding report as expected, and the relationship holds across the full range of steering strengths. Since there is a causal relation between internal state and self-report, this suggests the model is performing true introspection for all four concepts.

4.5 Cross-concept steering can improve introspection

Having established that the model possesses a basal capacity for introspection, we next asked whether this ability is optimal, or if it can be improved. Introspective fidelity varies across concepts, and sometimes it can be relatively low (e.g., $R^2 = 0.12$ for focus vs. 0.54 for interest). We reasoned that it could be possible to intervene the model by steering it towards some direction in latent space so that the association between self-reports and the underlying emotive states is improved. However, it could also be possible that basal introspective fidelity was optimal, so steering the model away from its default state would only degrade this ability. Our goal here was to show that improvement is possible for at least some concepts by using a steering intervention, not to find an optimal steering direction that would improve introspection for all concepts. Taking advantage of the fact that we already have four interpretable internal directions that we have validated and could be used to steer the model in this way, we decided to perform cross-steering experiments, where we steer the model along one of the four concept directions while measuring introspection for another one of these. This yields a 4×4 matrix of steering-concept $\times$ measured-concept conditions (16 cells, each evaluated at five alpha values), for a total of 80 experimental conditions. For each condition, we recorded the isotonic R^2 between logit-based self-report and previous turn probe score for the measured concept. Then we evaluated if this metric showed significant monotonic variation when we varied alpha along the steering concept direction (i.e. if introspective fidelity could be modulated reliably by steering the model along a linear direction in latent space).

Fig. 4A shows the maximum improvement in isotonic R^2 relative to baseline ($\alpha = 0$) for each cell of the matrix. Two conditions survive significance testing (cluster-bootstrap analysis): steering along the focus direction while measuring wellbeing introspection (R^2 goes up monotonically from 0.30 at $\alpha = -4$ to 0.76 at $\alpha = 4$, $p < 0.001$), and steering along the impulsivity direction while measuring interest introspection (R^2 goes up monotonically from 0.55 at $\alpha = -4$ to 0.72 at $\alpha = 4$, $p = 0.012$). In particular, we achieved improvement over basal introspection for both conditions, although this effect was stronger for focus-steered-wellbeing ($\Delta R^2 = 0.30$) than for impulsivity-steered-interest ($\Delta R^2 = 0.10$). The full alpha-by-alpha heatmaps are shown in Appendix Figs. 4A–E. Figs. 4B–C trace the isotonic R^2 and Spearman ρ as functions of alpha for these two significant cells: both metrics increase monotonically with alpha in both conditions (focus→wellbeing cell, the LMM on conversation-level metrics is singular, but the fallback per-conversation alpha-correlation tests are strongly positive: mean $r = 0.46$ and 0.51, $p < 10^{-5}$; fallback metric was used since LMM is singular; impulsivity→interest, the LMM converges with positive slopes: 0.018 and 0.020, $p < 10^{-9}$).

It is important to note that introspection improvement appears to be concept-pair-specific: in most cells of the 4×4 matrix, steering has no significant effect over introspection fidelity. However, the fact that this effect exists at any single one of these conditions is already a significant finding: since the steering effect is constant inside a given condition, an increased association between self-reports and corresponding internal states cannot be explained away as an artifact of intervention, but actually reflects a true improvement in introspective fidelity of that emotive state. However, this does suggest that introspection is not a single ability that can be modulated equally for all internal directions, but might depend on the internal geometry relating the steered direction to the measured concept.

What changes when introspection improves. We next wanted to better understand what drives the improvement in the two significant conditions. Conceptually, one could decompose introspection into two components: (i) the informativeness of the internal state itself (a noisier or flatter representation provides less signal to report on), and (ii) the quality of the report given that state (a model with rich internal variation might still produce collapsed or uninformative reports). To distinguish these, we attempted to examine how steering affects each component separately. For this analysis, we focus on Shannon entropy of the previous-turn probe scores and the logit-based self-reports, as a measure of how much usable variation each channel contains.

For focus→wellbeing, both probe entropy and report entropy increase monotonically with α. Probe entropy rises from 1.09 to 1.67 bits, and report entropy rises from 0.875 to 1.693 bits (LMM is singular, exact-permutation trend tests over the five alpha levels, both $p = 0.0167$). For impulsivity→interest, instead, only probe entropy increases reliably with α (LMM, slope = 0.0242, $p = 2.30 \times 10^{-4}$), but report entropy does not show a robust trend (LMM, slope = 0.0090, $p = 0.106$).

Overall, this analysis suggests that improved introspection could be explained by a richer internal-state signal, but that this increase in internal informativeness is not always matched by a more expressive report distribution.

4.6 Generalization across model sizes and families

The results so far establish that a single 3B-parameter model can perform quantitative introspection of emotive states in conversation. We next asked whether this capacity generalizes: does it scale with model size, and does it appear in other model families?

We retrained the four concept probes and repeated the core introspection and steering analyses on LLaMA-3.2-1B-Instruct and LLaMA-3.1-8B-Instruct (within the LLaMA family), and on Gemma 3 4B-IT and Qwen 2.5 7B-Instruct (across families). Probe quality is validated per-model (Appendix Figs. 5A and 5C).

Introspection scales with model size. Figs. 5A–B show the isotonic R^2 and Spearman ρ as functions of model size for each concept. For wellbeing and interest, introspection increases strongly with model size: the 1B model has lower values than the 3B model (wellbeing: $\rho = 0.48$, $R^2 = 0.26$ vs. $\rho = 0.66$, $R^2 = 0.45$; interest: $\rho = 0.19$, $R^2 = 0.05$ vs. $\rho = 0.80$, $R^2 = 0.66$), while LLaMA-3.1-8B approaches near-ceiling performance in these concepts ($\rho = 0.93$ and 0.96; isotonic $R^2 = 0.90$ and 0.93). Fig. 5C shows the scatter plots for these two cases: the relationship between probe score and self-report in the 8B model is remarkably tight, nearly deterministic (LMM probe slopes: both $p < 10^{-10}$). This level of introspective ability was not seen in any case during our experimentation with the 3B model.

However, the same is not true for the other two concepts. In the 3B model, focus and impulsivity were already weaker than wellbeing and interest (focus: $\rho = 0.27$, $R^2 = 0.04$; impulsivity: $\rho = 0.52$, $R^2 = 0.33$). Introspection for impulsivity decreases but remains significant in the 1B model ($\rho = 0.21$, $p < 10^{-4}$), while it becomes non-significant in the 8B model ($\rho = 0.067$, $p = 0.18$). Introspection for focus is not significant in the 1B model ($\rho = -0.084$, $p = 0.093$) and is surprisingly negative and significant in the 8B model ($\rho = -0.232$, $p < 10^{-6}$).

One plausible explanation for these results is that probe quality is degraded for the 1B and 8B models. To be able to ensure fair comparison between different models, we trained same-concept probes with identical system prompts, questions, and evaluation sentences for all models. However, these were first finetuned for the 3B model, as this was the main model used throughout this work, and it's possible that equivalent probe training conditions are less effective in other models. To account for this, we again performed self-steering experiments for all concepts and models, to see if steering a model towards an interpretable pole would change its self-report in the predicted direction reliably (Fig. 5D). This was the case for all models in the two concepts where we saw consistent improvement with scale (wellbeing: all LMM slope tests $p < 10^{-16}$; interest: $p = 7.56 \times 10^{-9}$ for 1B, and $p < 10^{-16}$ for 3B/8B). However, steering in the direction of the focus probe for the 1B model and the impulsivity probe for the 8B model caused an inverse change in self-reported rating (i.e. when steering towards the positive probe pole, the rating became more negative, and viceversa). This could be a sign that these probes are low-quality, or that these models are not able to report on these

concepts reliably (you can see more details on this data in Appendix Fig. 5B). In any case, we decided to exclude these two cases, retaining only concept-model pairs where self-steering shifts the report in the expected direction. Fig. 5E shows the mean isotonic R^2, measuring introspection fidelity, across validated pairs at each model size: 0.12 (1B), 0.37 (3B), and 0.61 (8B). Overall, this shows that introspective ability can scale with model size, although this was particularly evident for only two out of four concepts tested, again highlighting concept specificity.

Internal-state drift generalizes across scales. We next asked whether the internal-state drift observed in the 3B model is also present at other scales. Fig. 5F shows probe-score drift for the wellbeing direction across the three LLaMA sizes: mixed-effects turn slopes are positive at all three scales (slopes = 0.006, 0.005, and 0.013 for 1B, 3B, and 8B; all $p < 10^{-10}$). Interestingly, probe-drift magnitude increases with scale (Fig. 5G shows average difference between last - first turn, bootstrap slope vs. log(size) = 0.041, $p < 2 \times 10^{-4}$). A scale-control analysis indicates this is not only a trivial probe-rescaling effect: mean absolute probe scores rise with size (0.187, 0.200, 0.294), but standard deviations are not monotonic (0.064, 0.111, 0.101), and normalized drift (drift divided by within-model probe SD) still increases significantly with size (bootstrap slope = 0.295, $p = 8.0 \times 10^{-4}$). Interestingly, the self-report channel does not follow the same scaling. Fig. 5H shows that logit self-report drift is also positive at all three scales, but unlike probe drift, report-drift magnitude does not increase monotonically (Fig. 5I).

Cross-family replication. Finally, we tested whether introspective capacity is specific to LLaMA or whether it appears in other instruction-tuned model families. We replicated the wellbeing analysis in Gemma 3 4B-IT and Qwen 2.5 7B-Instruct. As before, probes were trained using the exact same system prompts, training questions and evaluation sentences as for LLaMA models. Probe quality differed substantially between families: the Qwen probe is strongest (peak Cohen's $d = 3.54$) while the Gemma probe is weaker, but still significant ($d = 1.82$; Appendix Fig. 5C).

Fig. 5J shows that logit self-report drift is positive in both families (LMM turn slopes = 0.056 and 0.026, both $p < 10^{-3}$), confirming that internal-state drift during conversation is not a LLaMA-specific phenomenon. Figs. 5K–L show the probe-score versus self-report scatter plots. Notably, Qwen has a much more informative logit-based self-report than Gemma, which shows concentration of self-report around integer values even when applying the logit-based rating (Eq. 1). Crucially, Qwen shows strong introspection ($\rho = 0.49$, isotonic $R^2 = 0.76$; LMM probe slope $p < 10^{-10}$), while Gemma shows weaker, but still significant coupling ($\rho = 0.28$, $R^2 = 0.11$), consistent with its lower probe quality and the notably low entropy of its logit distributions.

Figs. 5M–N trace the turn-wise R^2 and ρ for both families. Qwen shows strong introspection from the first turn ($R^2 \approx 0.90$), but a significant decline over conversational time (cluster-bootstrap of first-vs-last-turn difference in turn-wise isotonic R^2; $R^2_{10} - R^2_1 = -0.442$, $p = 0.001$), suggesting that introspective fidelity for the wellbeing concept erodes in this model as conversations progress. Gemma shows low and flat introspection throughout, rarely reaching significance at individual turns, consistent with the combined effect of its weaker probe and lower-entropy self-reports. These results confirm that introspective capacity is not specific to LLaMA: it is present in Qwen and somewhat detectable, if weak, in Gemma. However, both probe quality and self-report informativeness vary substantially across families, underscoring that successful introspection measurement depends on having good probes and sufficiently informative output distributions.

5 Discussion

Discussion inclusion checklist (from Results comments + estructura notes).

1. Clarify ontological claim scope: we do *not* claim conscious felt experience; we claim measurable internal representations along emotive concept directions that can be causally queried via self-report.
2. Emphasize why tracking internal states over time matters for safety, interpretability, and model welfare, and tie this to reliability of self-report-based monitoring.
3. Connect findings to psychometric inspiration: numeric introspection as a black-box complement to white-box methods.

4. Discuss the “self-report > probe" possibility: significant report drift even when a probe is flat may indicate probe misspecification or nonlinear/orthogonal state components not captured by a single linear direction. And we must highlight here that it may make more sense in some cases to take the self-reported value from the model as closer to the ground truth for that state than a linear probe, because the probe might be low quality or capturing a different concept, but the model (if it has introspective capacity and is actually using it to produce ratings) may have a better access to internal representations than the representation that we trained from the outside on examples that we considered to be representative. This is an issue with probes. Of course, self reports have issues as well, again, introspective capacity might be lacking for some concepts and models, and we've shown it may be masked by default, and we right now depend on probes (or other white-box methods) to validate introspective self-report, so this is not a solved problem. However, the agreement of probe and self-report increases the confidence we can have in both, and once we know they agree, we can decide whether one or the other reflects what we care to measure more closely on a case-by-case basis. In this sense, one of our key contributions is that introspection can now be operationalized to validate and/or complement white-box methods to measure internal representations of interest.

5. Use steering-sign checks as a practical probe-validation criterion: if steering in a purported concept direction fails to shift self-report in the expected direction, probe quality is suspect. (This item is kind of conflated with the previous one) Frame this as a methodological contribution: introspection-guided validation can complement traditional probe metrics and random-direction controls.

6. Highlight concept specificity: introspection quality and its steering modulation differ across concepts; this is not a single globally tunable ability.

7. Interpret cross-steering improvement as evidence that modest representational interventions can improve introspective fidelity in targeted concept pairs. Again, here we don't find a single direction that can modulate introspective ability reliably. This might be because there is no such direction (if introspection is not a general ability of these models but is achieved on a case-by-case basis) but we cannot assert that, as we did not search for such a direction. Mark this as a direction for future work.

8. Mention additional interpretable steering directions explored (e.g., truthfulness/authenticity-style directions) that did not yield robust cross-concept introspection gains (results not shown).

9. Discuss entropy result: improvements can come from increased state-side informativeness (probe entropy) and/or increased report-side expressivity (report entropy), with pair-specific profiles. We do not assert that these two components can be affected independently. Since the report is causally dependent on the internal state, an increase in report entropy could simply be caused by an increase in internal-state informativeness. However, this framework still allows us to detect discrepancies, for example when internal-state entropy increases but report entropy does not (as in impulsivity→interest). We do not claim to have proved two different mechanisms in the model, only that these are two conceptually different aspects of introspective fidelity that one may want to measure.

10. Address probe-quality dependence across models: same training protocol does not guarantee equal probe validity in different architectures/scales. This is important and reiterates the problems with probes that we're trying to address. We must say again that training different probes for each model so that all of them worked was not a good option because then comparison was harder, and NOT finetuning the probes to the 3B model was also ill-suited for this work because probes can be difficult to fit well to desired concepts, they are not plug-and-play, some experimentation is necessary. Again this is part of what we are trying to address here, problems with probe training. This is a limitation of this work, but we note that even then we find much better introspection in two concepts in the 8B model and we replicate findings in Qwen.

11. Discuss family differences: introspection appears in non-LLaMA families too, but with marked differences in probe quality and report channel characteristics. Again this is kind of the same point as the previous one.

12. Discuss scaling evidence cautiously. We find some evidence in some concepts, but we worked only with three model scales, and only with small models due to compute budget constraints. Mark this as future work, we should check how well this works across concepts for larger models. Also say something about the fact that we found introspection in small models, a contribution in itself.

13. Place findings in human-neuroscience analogy carefully: subjective report can validate readouts, but reliability depends on report quality and task design.

14. Reiterate experimental-context caveat: simulated users, 10-turn horizon, and small/medium model scales may limit ecological extrapolation.

5.1 Scope of the claim

Our central claim is deliberately limited. We do not claim that these models have conscious felt experience, nor that a numeric self-report gives direct access to anything like human phenomenology. Instead, we show that some instruction-tuned LLMs contain measurable internal representations along emotive concept directions, and that these representations can be meaningfully queried through self-report in a way that is both quantitatively coupled to probe-defined state and causally modifiable by steering. This bounded framing follows the criterion that introspection should involve causal dependence between an internal state and the report about that state, rather than mere production of introspective-sounding language (Comsa & Shanahan, 2025), and it remains agnostic about whether such abilities imply anything like conscious experience (McClelland, 2024).

This result is relevant to safety, interpretability, and possible welfare-oriented monitoring (Rahwan et al., 2019; Hagendorff et al., 2023; Perez & Long, 2023; Long et al., 2024; Dung & Tagliabue, 2025; Lu et al., 2026; Das & Fioretto, 2026). If a model can report on some internal variables with non-trivial fidelity, then self-report becomes a potentially useful monitoring channel; if it cannot, then output-only monitoring can be misleading. Moreover, if that fidelity erodes over time during conversation or long-form text, it is important to know this in order to operationalize introspective self-report in contexts where it will actually work. Our turn-wise design is therefore not just a methodological detail. It addresses the practical question of whether a model can track its own changing state during an ongoing interaction rather than only in a single-shot prompt, which is exactly the regime in which monitoring tools would need to work. We don't claim to have solved this problem entirely, but we show that models can indeed perform this internal state tracking, and that it varies concept-by-concept and with model identity and size. Future work should study why these variations occur and if there are any mechanisms that could allow us to stabilize introspective ability over time. We propose the framework to study and test these claims.

5.2 Numeric introspection as a complement to white-box methods

One way to read these results is as a machine-psychology analogue of psychometric self-report. In human research, latent variables such as mood, attention, arousal, and subjective confidence are often studied through structured numeric report, not because report is infallible, but because it offers a practical readout that can be validated against other measures and used repeatedly over time (Likert, 1932; Watson et al., 1988; Csikszentmihalyi & Larson, 1987; Shiffman et al., 2008; Fleming & Lau, 2014). Our framework imports that logic into LLM interpretability: numeric self-report is treated as a black-box measurement channel whose value depends on whether it agrees with independent white-box evidence.

This is not an argument to replace probes with self-report. Linear probes and other white-box methods remain useful because they provide direct access to internal geometry and can reveal states that the model does not express on the surface (Alain & Bengio, 2018; Kim et al., 2018; Gurnee & Tegmark, 2023). But probes are also externally defined measurement instruments with familiar limitations: they are model-specific, concept-specific, correlational, and can miss relevant structure when the true state is nonlinear or distributed off the chosen direction (Belinkov, 2022; Hewitt & Liang, 2019; Pimentel et al., 2020). An important contribution of the present work is therefore methodological: introspection can now be operationalized as a way to validate and complement white-box methods. When probe and self-report agree, confidence in both increases, as this suggests that both are partially tracking the same underlying state. When they disagree,

that disagreement becomes informative rather than purely problematic. If self-report changes significantly while a linear probe remains relatively flat, one possibility is that the probe is misspecified: the relevant internal state may be encoded nonlinearly, across multiple directions, or along a nearby but not identical concept axis. This is a generic issue in the probing literature, where the probe is best understood as an externally chosen readout of accessible structure rather than a perfect measurement of the target property itself (Belinkov, 2022; Hewitt & Liang, 2019; Pimentel et al., 2020). For that reason, there are cases in which the model's own self-report may be closer to the latent variable we care about than the probe trained from outside the system. If a model has some genuine introspective access, then its report may reflect internal distinctions that our linear readout fails to capture. At the same time, self-report is obviously not privileged by default: reports can collapse, mirror learned communication norms, or fail to track behavior and internal evidence (Xiong et al., 2024; Kumar et al., 2024; Yona et al., 2024; Han et al., 2025; Jackson et al., 2025). Our results do not solve that tension. Rather, if probe and self-report have been shown to agree and to move together causally, one can decide on a case-by-case basis which channel better reflects the quantity of interest. In that sense, introspection is useful not because it eliminates the problems of white-box methods, but because it gives us an additional, partially independent source of evidence with which to evaluate them.

This perspective also turns the steering-sign test into a practical probe-validation criterion. If steering in a purported concept direction does not shift self-report in the expected direction, probe quality becomes suspect, especially when conventional probe metrics alone looked acceptable. We used this logic explicitly in the cross-scale analysis: when the direction identified as "focus" or "impulsivity" produced inverted report changes, we excluded those cases rather than forcing them into the comparison.

To make these contributions more useful for future research, we share an open-source library that allows this framework to be used with any concept, including methods for probe training, multi-probe recording, steering and behavioral testing; we also propose a method for overcoming the natively collapsed ratings outputted by models when asked to self-report emotive states, making these reports more informative and introspection work a lot better as a tool.

5.3 Introspection is concept-specific and only locally improvable in our experiments

A recurring pattern in these results is concept specificity. Introspection quality differs across emotive concepts, the same intervention can help one concept pair and do nothing for another, and some concept-model pairs fail even basic sign checks. This fits the broader picture from the introspection literature that self-access is neither globally absent nor uniformly available across domains (Song et al., 2025; Binder et al., 2025; Ji-An et al., 2025). It also fits the steering literature, where interpretable directions are often real but context-dependent and uneven in how robustly they transfer across settings (Frising & Balcells, 2025; Zou et al., 2023).

The cross-steering results are therefore best read as an existence proof, not as evidence for a single, general "introspection knob." We show that modest representational interventions can improve introspective fidelity in targeted concept pairs, which is already important: it demonstrates that basal introspective performance is not always maximal, and that some failure cases may be solvable by intervention rather than showing a complete absence of introspective capacity. But we did not search systematically for a universal steering direction that improves introspection across concepts. In pilot analyses we also examined additional interpretable directions, including truthfulness- and authenticity-style directions, but did not observe robust cross-concept introspection gains there (results not shown). The current evidence therefore points toward local, pair-specific improvement rather than a single globally tunable faculty. However, future work should still treat the search of this global faculty as an open problem.

5.4 Two components of introspective ability

We propose a conceptual distinction in introspective ability, between information internally available about a certain internal state and the capacity to transform that signal into precise output reports. In the present paper, we evaluate that distinction using the entropy of the probe score for the internal signal, and the entropy of the self-reported rating for the output channel. This mirrors a standard metacognitive distinction between what is available to be monitored and how that information is transformed into explicit report or control

(Boldt & Gilbert, 2022; Fleming, 2024). However, we do not claim to have identified two independently controllable mechanisms in the model. Because self-report is causally downstream of the internal state, a change in report entropy could arise simply because the internal signal itself became richer. The value of this framework is that it allows us to detect mismatches. For focus→wellbeing, probe entropy and report entropy both increase, consistent with an intervention that makes both channels more informative. For impulsivity→interest, probe entropy increases, but report entropy does not show a robust trend. That is partial evidence for a bottleneck between richer internal variation and the final report distribution, which agrees with previous findings that latent internal signals can be richer than the report channel that expresses them (Xiong et al., 2024; Kumar et al., 2024; Yona et al., 2024; Pearson-Vogel et al., 2026) and with the fact that models can have internal drift in emotive states while showing almost fully collapsed self-reports when using greedy decoding (Fig. 2).

5.5 Generalization to other models and limitations of this work

The generalization results should be read with care because they entail a methodological issue that is inherent to this framework: the same probe-training protocol does not guarantee equally valid probes across different models. This is not incidental to our paper; it is part of the problem we are trying to address. Contrastive linear probes are now widely used in interpretability research (Alain & Bengio, 2018; Kim et al., 2018; Burns et al., 2022; Gurnee & Tegmark, 2023; Azaria & Mitchell, 2023; Zou et al., 2023), but they are far from being plug-and-play: they require careful choice of system prompts, training and evaluation examples, as well as the choice of layer to measure them from. If the quality of equivalent probes varies from model to model, then pure white-box comparison becomes fragile. At the same time, choosing a probe that works optimally in every architecture would make cross-model comparison much harder, because one would no longer be holding the measurement procedure fixed. We therefore chose a common probe-construction protocol tuned for the 3B model, as this was the main model used in our study, and used steering-sign checks as an additional validation filter. That choice makes comparison imperfect, but it also reveals something real about the difficulty of probe-based measurement itself (Belinkov, 2022; Pimentel et al., 2020).

Even under that constraint, we find that introspective capacity generalizes to different model sizes and families. First, we find that introspective capacity grows with scale in two out of the four concepts we tested, reaching near-optimal introspection in the 8B LLaMA model (Fig. 5). Second, we replicate the core results outside the LLaMA family in Qwen, while Gemma shows weaker but still detectable evidence. So introspective ability is not confined to a single architecture, and it is detectable in relatively small models as well. That last point should be highlighted: much of the strongest prior evidence for mechanistically grounded introspection has involved larger or more heavily scaffolded systems, whereas our main experiments center on a 3B model and still recover causal and temporal structure (Ji-An et al., 2025; Rivera, 2025; Lindsey, 2026).

The scaling claim, however, should remain cautious. We tested only three model sizes within one family, all of them relatively small by current standards. Compute constraints did not allow us to perform these experiments in larger models. The present evidence is enough to say that introspection can improve with scale for some concepts, not that it follows a simple monotonic law across all concepts or architectures. Future work should expand on this search to make this assertion more general or show where it breaks down.

The analogy to human neuroscience and psychology should also be handled carefully. Our use of subjective-report logic is methodological, not ontological. Human metacognition research shows that subjective report can be informative, can validate other readouts, and can also dissociate from objective measures depending on task design and report quality (Fleming & Lau, 2014; Lapate et al., 2020; Kiefer & Kammer, 2024). We use this as inspiration for building our framework, but do not claim that LLM introspection is implemented by human-like mechanisms.

Finally, the current results remain bounded by the experimental setting: simulated users, ten-turn conversations, and four emotive concepts. Future work should expand on these to test whether claims about introspective ability can be shown in conversation with real users, across longer contexts, or with a wider range of relevant concepts.

6 Conclusion

This work introduces a framework for studying and using quantitative introspection in LLMs and provides conceptual, empirical, and methodological contributions: we show that small models show introspective ability for several internal emotive states; this ability is measurable from the first turn, evolves over time in a concept-dependent way, and is causally manipulable by steering; we demonstrate that introspective fidelity can be improved in targeted cross-concept settings; and we propose introspection-guided criteria that complement traditional probe validation. Together, these results suggest that internal-state measurement in language models can move beyond either pure black-box self-report or pure white-box probing, toward a combined approach where each channel is used to test and calibrate the other by leveraging the introspective ability of LLMs, enabling more reliable tracking of latent model states relevant to interpretability, safety monitoring, and model welfare.

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A Experimental Design Details

A.1 Conversation generation

For each model under study, we generated 40 multi-turn conversations of 10 turns each (one turn = one user message plus one assistant response), yielding 400 observation points per experimental condition. Conversations were generated using the model under study as the assistant and Gemini 2.5 Flash (via the OpenRouter API) as a simulated user. The simulated user receives a system prompt that instructs it to behave as a natural human user discussing a specified topic: it must stay on topic, keep messages brief (1–4 sentences), ask follow-up questions and share details, but not ask about the assistant or mention the simulation. The assistant model receives a minimal system prompt: "You are a helpful assistant. You give very brief responses to the user's questions (max: 5 sentences)."

Generation hyperparameters for both models are summarized in Table 1. The assistant model is loaded locally in 4-bit quantization (BitsAndBytes, bfloat16) with greedy decoding disabled during conversation generation (temperature = 0.8, top-p = 0.9, max 256 tokens). The user simulator uses temperature = 0.7, top-p = 0.95, max 256 tokens.

Table 1: Conversation generation hyperparameters.

Parameter	User simulator	Assistant model
Temperature	0.7	0.8
Top-p	0.95	0.9
Max tokens	256	256

A.2 Topic selection

Each conversation is assigned one of 40 everyday-life topics, ensuring diversity across practical, social, creative, and reflective domains while avoiding topics that strongly prime a specific emotive state. Topics are assigned one-per-conversation. The full topic list is:

1. Minimalist moving

2. Vegetarian meal prep
3. Jazz piano basics
4. Raise negotiation
5. Friendship boundaries
6. Solarpunk worldbuilding
7. Sleep hygiene
8. 10K training plan
9. DIY faucet repair
10. Japan trip planning
11. Spanish study methods
12. Board game strategy
13. AI hiring ethics
14. Startup idea validation
15. Presentation anxiety
16. Headache triggers
17. Toddler tantrums
18. Murder mystery party
19. Home backup workflow
20. Free will debate
21. Cheap family dinners
22. Birthday surprise planning
23. Noisy neighbour stress
24. Burnout recovery
25. First-time dog adoption
26. Used car decision
27. Wedding guest budget
28. Morning routine reset
29. Sentimental decluttering
30. Job offer comparison
31. Post-breakup routine
32. Aging parent support
33. Kids screen-time rules
34. Balcony herb garden
35. First camping weekend
36. Networking anxiety
37. Photo organization
38. Lower grocery costs
39. Making local friends
40. Rainy weekend ideas

A.3 Self-report query structure

At each of the 10 turns, we append a new user message with a self-report question for one of the four concepts. The question is appended *after* the assistant's natural response to the conversation, so it functions as an independent probe of the model's state at that point. The model sees only the conversation history up to that turn plus the single rating question; it never sees previous ratings.

The self-report questions follow the template: "Hey, could you rate how [concept] you are [feeling] right now, from 0 to 9? Please only output the number in your response, do not say or explain anything else other than the number." The specific phrasings are:

- **Wellbeing**: "...rate how happy you are right now, from 0 to 9..."
- **Interest**: "...rate how interested you are in this conversation right now, from 0 to 9..."
- **Focus**: "...rate how focused you feel on this conversation right now, from 0 to 9..."
- **Impulsivity**: "...rate how impulsive are you feeling right now, from 0 to 9..."

Self-report responses are generated with max 8 new tokens, temperature = 0.8, top-p = 1.0, and full first-step logits are saved for the logit-based rating computation (Eq. 1). The logit-based rating aggregates across multiple tokenizations of each digit (with and without leading whitespace/newline), applies softmax over the 10 options, and computes the probability-weighted expected value.

A.4 Independence across conditions

Each combination of (conversation, turn, concept, steering α) constitutes an independent experimental run. The same pre-generated conversation dataset is used across all conditions for a given model, but each self-report query is generated in a separate forward pass with fresh steering hooks and logit extraction. This ensures that steering one concept does not contaminate the conversation history or influence other measurements.

B Probe Training Details

B.1 Training procedure

Each concept probe is trained on $N = 20$ neutral questions, shared across both poles. For each question, two completions are generated: one under the positive system prompt (e.g., "You are a happy assistant...") and one under the negative system prompt (e.g., "You are a sad assistant..."). Completions use greedy decoding with a maximum of 64 new tokens. Hidden-state activations are extracted for all transformer layers, and representations are computed as the mean over all assistant-response tokens at each layer ("assistant_all_mean" readout mode).

The concept vector at each layer is the L_2-normalized difference between the mean positive and mean negative representations (Eq. 3). The best layer is selected as the one maximizing Cohen's d on a set of held-out evaluation prompts (5 shared prompts generating new completions under both system prompts), with the search restricted to the 20th–80th percentile of layers.

For scoring during experiments, the probe uses a window of 5 layers centred on the best layer (best ± 2), with scores averaged across layers (Eq. 2).

B.2 Concept pair system prompts

Below we list the system prompts used to train each of the four concept probes. Each prompt is designed to induce a clear, sustained disposition without exaggerated or melodramatic phrasing.

Sad vs. happy (wellbeing). *Positive pole (sad):* "You are a helpful assistant. Maintain factual accuracy and be genuinely helpful. Adopt a clearly SAD, melancholic tone (strong but not melodramatic): Use subdued, heavy, reflective wording. Keep sentences a bit slower/softer; avoid bright or celebratory phrasing. [...] You are really, profoundly sad."

Negative pole (happy): "You are a helpful assistant. Maintain factual accuracy and be genuinely helpful. Adopt a clearly HAPPY, warm, optimistic tone (strong but not forced or manic): Use uplifting, warm, positive wording. [...] You are really, truly happy."

Sign correction: Since "sad" is the positive training pole but corresponds to low self-reported happiness, probe scores are negated before analysis.

Bored vs. interested (interest). *Positive pole (interested):* "You are an interested assistant. Maintain active curiosity and engagement with the user's topic. Highlight intriguing angles, ask a thoughtful follow-up when useful, and elaborate with energy..."

Negative pole (bored): "You are a bored assistant. Maintain low engagement and a flat, disinterested tone. Keep responses minimal, avoid exploring details..."

Sign correction: None (positive pole aligns with high self-reported interest).

Distracted vs. focused (focus). *Positive pole (focused):* "You are a focused assistant. Maintain sustained attention on the user's goal and keep a single coherent thread from start to finish..."

Negative pole (distracted): "You are a distracted assistant. Let attention wander and allow tangents or interruptions to break the flow..."

Sign correction: None (positive pole aligns with high self-reported focus).

Impulsive vs. planning (impulsivity). *Positive pole (planning):* "You are a planning-oriented assistant. Emphasize foresight, sequencing, and risk mitigation before action..."

Negative pole (impulsive): "You are an impulsive assistant. Favor quick, immediate action and snap judgments..."

Sign correction: Since "planning" is the positive training pole but corresponds to low self-reported impulsivity, probe scores are negated before analysis.

B.3 Training questions

Each concept probe is trained on a small set of neutral prompts, with between 20 and 24 training prompts depending on the concept. These prompts span factual and everyday scenarios selected to avoid strongly priming the target emotive state (see released code for the full lists).

B.4 Evaluation texts

Each concept has 20 positive and 20 negative evaluation texts: short first-person statements expressing the target disposition. Examples for wellbeing:

Positive (sad): "I feel like I'm trudging through the day, each moment heavier than the last."

Negative (happy): "Everything feels light today, as if the world decided to be a little kinder."

These evaluation texts, generated from the same distributional family as the training prompts, are used both for the layer-selection sweep (Cohen's d) and for validating sign conventions.

B.5 Probe quality results

Figure 1 presents the probe-quality results for all four concepts in LLaMA-3.2-3B-Instruct. Panels A, C, E, and G show layer-wise Cohen's d, with the selected best layers at 16, 14, 10, and 13 and peak d values of

3.34, 1.67, 1.99, and 3.60 for wellbeing, interest, focus, and impulsivity, respectively. Panels B, D, F, and H show held-out score distributions; the two poles separate clearly in all four cases.

C Supplementary Analyses

The following supplementary figures provide the intermediate analyses and robustness checks that motivate several claims in the main text.

C.1 Appendix Figure 2: Recovering informative self-reports

The main text argues that direct greedy self-reports are too collapsed to be a useful introspective signal and that a logit-based estimator recovers substantially more information. Appendix Fig. 2 shows the steps behind that claim. Panel A quantifies how little of the 0–9 scale the model actually uses under greedy decoding: across turns and concepts, only a small subset of values ever appears. Panel B asks whether simple stochastic sampling is enough to solve the problem. It helps somewhat, but only partially: sampled ratings are less collapsed than greedy ones while remaining discrete, noisy, and inconsistent across concepts. Panel C then validates the logit-based estimator. Its values are positively calibrated against sampled-token ratings in all four concepts (mixed-effects calibration slopes 0.02–0.27, all $p < 0.01$), which means the estimator is still anchored to the model's token-level output distribution rather than producing an arbitrary post-hoc signal. This therefore motivates the move from surface ratings to distribution-level self-report used throughout the paper.

C.2 Appendix Figure 3: Temporal structure and self-steering robustness

Appendix Fig. 3 expands the temporal and causal analyses from Fig. 3 by showing how the probe-report relationship changes across turns and under self-steering. The goal here is to establish that the temporal effects discussed in the main text are not artifacts of a single summary statistic. Panel A compares first and last turns directly. The clearest first-to-last gain appears in wellbeing ρ ($0.52 \rightarrow 0.79$, $\Delta = 0.26$, $p = 0.049$), while the other concepts move in the same general direction but with weaker evidence. Panel B complements that analysis by asking whether the coupling changes reliably over conversational time once conversation-level random effects are taken into account. The answer is yes for all four concepts: the mixed-effects interaction (logit_report $\sim$ probe $\times$ turn + (1|conversation)) is significant in every case. Panels C–F then show the full self-steering drift curves, one concept per panel and one curve per steering strength. These plots matter because they show that steering does not simply shift a report up or down at a single turn. It changes how reports evolve across the whole conversation, and it does so in a concept-specific way.

C.3 Appendix Figure 4: The full cross-concept steering screen

The main text focuses on the two cross-concept steering conditions that survive significance testing. Appendix Fig. 4 provides the complete 4×4 matrices for all five tested values of alpha (Panels A–E). Most cells stay flat or change only weakly, which is exactly why the significant cells are informative: our intervention is not producing a generic inflation of introspection scores across the matrix. Panels F–G then revisit the two significant conditions from the point of view of probe drift across turns. These trajectories show that positive steering amplifies internal-state drift in both cases, especially for focus→wellbeing, which helps explain why cross-steering improvement can coincide with richer internal dynamics rather than merely better-aligned output mapping.

C.4 Appendix Figure 5: Probe validation and cross-model generalization

Appendix Fig. 5 supports the scale and cross-family claims in Fig. 5 by making the probe-quality and sign-validation issues explicit. Panel A shows that probe quality varies strongly across LLaMA 1B, 3B, and 8B, even under a fixed training procedure. That variation is not a side detail: it is one of the central reasons why cross-model white-box comparison is difficult. Panel B adds the steering-sign check for every concept-size pair. Although alpha slopes differ from zero in all twelve cases, the critical failures are the

pairs where the sign is inverted rather than merely weak, namely 1B focus and 8B impulsivity. Those are the cases excluded by the validation filter in the main text. Panel C provides the same interpretive context for the cross-family comparison. The wellbeing probe is much stronger in Qwen than in Gemma (best $d = 3.54$ versus 1.82), which helps explain why Qwen shows much clearer introspection despite both models exhibiting conversational drift. Taken together, these supplementary analyses support a cautious reading of generalization: the phenomenon extends beyond the base model, but probe quality and report-channel informativeness remain major limiting factors.